Ayush Tewari

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Academic Appointments

2025–present **University of Cambridge.**
Assistant Professor, Department of Engineering

2026–present **St. John's College, Cambridge.**
Fellow and College Lecturer

2021–2024 **Massachusetts Institute of Technology.**
Postdoctoral Researcher with Profs. William Freeman, Joshua Tenenbaum, and Vincent Sitzmann

2021 **Max Planck Institute for Informatics.**
Postdoctoral Researcher with Prof. Christian Theobalt

2020 **Stanford University.**
Research Intern with Prof. Maneesh Agrawala

Education

2016–2021 **Max Planck Institute for Informatics and Saarland University.**
Doctor of Engineering (Dr.-Ing.), summa cum laude
Thesis: "Self-Supervised Reconstruction and Synthesis of Faces"
Supervisor: Prof. Christian Theobalt

2014–2015 **INRIA and Grenoble Institute of Technology.**
Master of Science in Computer Science
Thesis: "Image Blending using Local Phase"
Supervisors: Dr. George Drettakis and Dr. Adrien Bousseau

2010–2014 **International Institute of Information Technology Hyderabad.**
Bachelor of Technology (Honours) in Computer Science

Selected Honours and Awards

2026 NVIDIA Academic Grant

2023 NeurIPS Spotlight for "Diffusion with Forward Models: Solving Stochastic Inverse Problems without Direct Supervision", awarded to top ∼3% submissions.

2022 Otto Hahn Medal by the Max Planck Society, awarded for outstanding scientific achievement by junior scientists.

2022 ECCV Oral for "Neural Radiance Transfer Fields for Relightable Novel-view Synthesis with Global Illumination", awarded to top ∼3% submissions.

2022 BMVC Best Paper Honorable Mention for "VoRF: Volumetric Relightable Faces".

2021 CVPR Oral for "i3DMM: Deep Implicit 3D Morphable Model of Human Heads", awarded to top ∼6% submissions.

2020 CVPR Oral for "StyleRig: Rigging StyleGAN for 3D Control over Portrait Images", awarded to top ∼6% submissions.

2019 CVPR Oral for "FML: Face Model Learning from Videos", awarded to top ∼6% submissions.

2018 CVPR Oral for "Self-supervised Multi-level Face Model Learning for Monocular Reconstruction at over 250 Hz", awarded to top ∼2% submissions.

2018 Invited paper for TPAMI special issue on the best of ICCV 2017 "High-Fidelity Monocular Face Reconstruction Based on an Unsupervised Model-Based Face Autoencoder".

2017 ICCV Oral for "MoFA: Model-based Deep Convolutional Face Autoencoder for Unsupervised Monocular Reconstruction", awarded to top ∼2% submissions.

Peer-reviewed Conference Papers

* equal first authors † equal advising

[C1] J. Gaubil, M. Stary, L. Martinez, A. Geiger, M. Ovsianikov†, A. Tewari†, and V. Sitzmann†. Understanding Multi-View Transformers. In *Neural Information Processing Systems (NeurIPS)*, 2026.

[C2] A. Kiruga, Z. Li, R. Cai, H. Lillemark, X. Wu, A. Tewari, and Y. Du. Unified Generative-Predictive Modeling for 4D Scene Understanding. In *Neural Information Processing Systems (NeurIPS)*, 2026.

[C3] L. Lyu, A. Tewari, J. Chen, T. Leimkühler, and C. Theobalt. Faster 3D Gaussian Splatting Convergence via Structure-Aware Densification. In *SIGGRAPH*, 2026.

[C4] F. O'Mahony, R. Cipolla, and A. Tewari. VDAWorld: World Modelling via VLM-Directed Abstraction and Simulation. In *SIGGRAPH Asia*, 2026.

[C5] C. Smith, D. Charatan, A. Tewari, and V. Sitzmann. Flowmap: High-quality camera poses, intrinsics, and depth via gradient descent. In *2025 International Conference on 3D Vision (3DV)*, pages 389–400. IEEE, 2025.

[C6] V. Gupta, R. S. V. Girish, A. Tewari, and K. Mitra. GAURA: Generalizable Approach for Unified Restoration and Rendering of Arbitrary Views. In *European Conference on Computer Vision (ECCV)*, 2024.

[C7] A. Harrington, V. DuTell, M. Hamilton, A. Tewari, S. Stent, W. T. Freeman, and R. Rosenholtz. COCO-Periph: Bridging the Gap Between Human and Machine Perception in the Periphery. In *The Twelfth International Conference on Learning Representations*, 2024.

[C8] L. Lyu, A. Tewari, M. Habermann, S. Saito, M. Zollhöfer, T. Leimkühler, and C. Theobalt. Manifold Sampling for Differentiable Uncertainty in Radiance Fields. In *SIGGRAPH Asia*, 2024.

[C9] V. van der Brugge, M. Pollefeys, J. B. Tenenbaum, K. M. Jatavallabhula, and A. Tewari. PickScan: Object discovery and reconstruction from handheld interactions. In *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 10167–10173. IEEE, 2024.

[C10] A. Tewari*, T. Yin*, G. Cazenavette, S. Rezchikov, J. B. Tenenbaum, F. Durand, W. T. Freeman, and V. Sitzmann. Diffusion with Forward Models: Solving Stochastic Inverse Problems Without Direct Supervision. In *Neural Information Processing Systems (NeurIPS) (Spotlight Presentation)*, 2023.

[C11] Y. Du, C. Smith, A. Tewari†, and V. Sitzmann†. Learning to Render Novel Views from Wide-Baseline Stereo Pairs. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.

[C12] K. Jatavallabhula, A. Kuwajerwala, Q. Gu, M. Omama, T. Chen, S. Li, G. Iyer, S. Saryazdi, N. Keetha, A. Tewari, J. Tenenbaum, C. de Melo, M. Krishna, L. Paull, F. Shkurti, and A. Torralba. ConceptFusion: Open-set Multimodal 3D Mapping. In *RSS*, 2023.

[C13] X. Pan, A. Tewari, T. Leimkühler, L. Liu, A. Meka, and C. Theobalt. Drag Your GAN: Interactive Point-based Manipulation on the Generative Image Manifold. In *ACM SIGGRAPH 2023 Conference Proceedings*, 2023.

[C14] C. Smith, Y. Du, A. Tewari, and V. Sitzmann. FlowCam: Training Generalizable 3D Radiance Fields without Camera Poses via Pixel-Aligned Scene Flow. In *Neural Information Processing Systems (NeurIPS)*, 2023.

[C15] M. B R, A. Tewari, X. Pan, M. Elgharib, and C. Theobalt. gCoRF: Generative Compositional Radiance Fields. In *International Conference on 3D Vision (3DV)*, 2022.

[C16] A. Tewari, M. B R, X. Pan, O. Fried, M. Agrawala, and C. Theobalt. Disentangled3D: Learning a 3D Generative Model with Disentangled Geometry and Appearance from Monocular Images. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2022.

[C17] A. Harrington, V. DuTell, A. Tewari, M. Hamilton, S. Stent, R. Rosenholtz, and W. T. Freeman. Exploring perceptual straightness in learned visual representations. In *The Eleventh International Conference on Learning Representations (ICLR)*, 2022.

[C18] L. Lyu, A. Tewari, T. Leimkuehler, M. Habermann, and C. Theobalt. Neural Radiance Transfer Fields for Relightable Novel-view Synthesis with Global Illumination. In *European Conference on Computer Vision (ECCV) (Oral Presentation)*, 2022.

[C19] X. Pan, A. Tewari, L. Liu, and C. Theobalt. GAN2X: Non-Lambertian Inverse Rendering of Image GANs. In *International Conference on 3D Vision (3DV)*, 2022.

[C20] P. Rao, M. B R, G. Fox, T. Weyrich, B. Bickel, H.-P. Seidel, H. Pfister, W. Matusik, A. Tewari, C. Theobalt, and M. Elgharib. VoRF: Volumetric Relightable Faces. In *British Machine Vision Conference (BMVC) (Oral Presentation, Best Paper Honorable Mention)*, 2022.

[C21] P. Sharma, A. Tewari, Y. Du, S. Zakharov, R. A. Ambrus, A. Gaidon, W. T. Freeman, F. Durand, J. B. Tenenbaum, and V. Sitzmann. Neural Groundplans: Persistent Neural Scene Representations from a Single Image. In *The Eleventh International Conference on Learning Representations (ICLR)*, 2022.

[C22] M. B R, A. Tewari, T.-H. Oh, T. Weyrich, B. Bickel, H.-P. Seidel, H. Pfister, W. Matusik, M. Elgharib, and C. Theobalt. Monocular Reconstruction of Neural Face Reflectance Fields. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.

[C23] M. B R, A. Tewari, H.-P. Seidel, M. Elgharib, and C. Theobalt. Learning Complete 3D Morphable Face Models from Images and Videos. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.

[C24] G. Fox, A. Tewari, M. Elgharib, and C. Theobalt. StyleVideoGAN: A Temporal Generative Model using a Pretrained StyleGAN. In *British Machine Vision Conference (BMVC) (Oral Presentation)*, 2021.

[C25] L. Lyu, M. Habermann, L. Liu, M. B. R, A. Tewari, and C. Theobalt. Efficient and Differentiable Shadow Computation for Inverse Problems. In *IEEE International Conference on Computer Vision (ICCV)*, 2021.

[C26] E. Tretschk, A. Tewari, V. Golyanik, M. Zollhöfer, C. Lassner, and C. Theobalt. Non-Rigid Neural Radiance Fields: Reconstruction and Novel View Synthesis of a Dynamic Scene From Monocular Video. In *IEEE International Conference on Computer Vision (ICCV)*, 2021.

[C27] T. Yenamandra, A. Tewari, F. Bernard, H.-P. Seidel, M. Elgharib, D. Cremers, and C. Theobalt. i3DMM: Deep Implicit 3D Morphable Model of Human Heads. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR) (Oral Presentation)*, 2021.

[C28] Y. Zhou, M. Habermann, I. Habibie, A. Tewari, C. Theobalt, and F. Xu. Monocular Real-time Full Body Capture with Inter-part Correlations. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.

[C29] A. Tewari, M. Elgharib, G. Bharaj, F. Bernard, H.-P. Seidel, P. Pérez, M. Zollhöfer, and C. Theobalt. StyleRig: Rigging StyleGAN for 3D Control over Portrait Images. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR) (Oral Presentation)*, 2020.

[C30] J. Thies, M. Elgharib, A. Tewari, C. Theobalt, and M. Nießner. Neural Voice Puppetry: Audio-driven Facial Reenactment. In *European Conference on Computer Vision (ECCV)*, 2020.

[C31] E. Tretschk, A. Tewari, V. Golyanik, M. Zollhöfer, C. Stoll, and C. Theobalt. PatchNets: Patch-Based Generalizable Deep Implicit 3D Shape Representations. In *European Conference on Computer Vision (ECCV)*, 2020.

[C32] E. Tretschk, A. Tewari, M. Zollhöfer, V. Golyanik, and C. Theobalt. DEMEA: Deep Mesh Autoencoders for Non-Rigidly Deforming Objects. In *European Conference on Computer Vision (ECCV) (Spotlight Presentation)*, 2020.

[C33] A. Tewari, F. Bernard, P. Garrido, G. Bharaj, M. Elgharib, H.-P. Seidel, P. Pérez, M. Zollhöfer, and C. Theobalt. FML: Face Model Learning from Videos. In *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR) (Oral Presentation)*, 2019.

[C34] A. Tewari, M. Zollhöfer, P. Garrido, F. Bernard, H. Kim, P. Pérez, and C. Theobalt. Self-supervised Multi-level Face Model Learning for Monocular Reconstruction at over 250 Hz. In *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR) (Oral Presentation)*, 2018.

[C35] H. Kim, M. Zollhöfer, A. Tewari, J. Thies, C. Richardt, and C. Theobalt. InverseFaceNet: Deep Single-Shot Inverse Face Rendering From A Single Image. In *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.

[C36] Q. Sun, A. Tewari, W. Xu, M. Fritz, C. Theobalt, and B. Schiele. A Hybrid Model for Identity Obfuscation by Face Replacement. In *European Conference on Computer Vision (ECCV)*, 2018.

[C37] A. Tewari, M. Zollhöfer, H. Kim, P. Garrido, F. Bernard, P. Perez, and C. Theobalt. MoFA: Model-based Deep Convolutional Face Autoencoder for Unsupervised Monocular Reconstruction. In *The IEEE International Conference on Computer Vision (ICCV) (Oral Presentation)*, 2017.

Peer-reviewed Journal Articles

* equal first authors † equal advising

[J1] H. Chen, K. Pandey, R. Wu, M. Gadelha, Y. Hold-Geoffroy, A. Tewari, N. J. Mitra, C. Zheng, and P. Guerrero. ViPS: Video-informed Pose Spaces for Auto-Rigged Meshes. *ACM Transactions on Graphics (SIGGRAPH Asia)*, 2026.

[J2] D. Shi, L. Jiang, K. M. Collins, S. Wu, A. Tewari, and M. Zilka. Perceived authenticity drives gaze behavior when watching AI-generated videos of physical scenes. *Scientific Reports*, 2026.

[J3] T. P. O'Connell, T. Bonnen, Y. Friedman, A. Tewari, V. Sitzmann, J. B. Tenenbaum, and N. Kanwisher. Approximating Human-Level 3D Visual Inferences With Deep Neural Networks. *Open Mind*, 9:305–324, 2025.

[J4] L. Lyu, A. Tewari, M. Habermann, S. Saito, M. Zollhöfer, T. Leimküehler, and C. Theobalt. Diffusion Posterior Illumination for Ambiguity-aware Inverse Rendering. *ACM Transactions on Graphics (Proceedings SIGGRAPH Asia)*, 2023.

[J5] M. Mendiratta, X. Pan, M. Elgharib, K. Teotia, M. B R, A. Tewari, V. Golyanik, A. Kortylewski, and C. Theobalt. AvatarStudio: Text-driven Editing of 3D Dynamic Human Head Avatars. *ACM Transactions on Graphics (Proceedings SIGGRAPH Asia)*, 2023.

[J6] A. Petitjean, Y. Poirier-Ginter, A. Tewari, G. Cordonnier, and G. Drettakis. ModalNeRF: Neural Modal Analysis and Synthesis for Free-Viewpoint Navigation in Dynamically Vibrating Scenes. *Computer Graphics Forum (EGSR)*, 2023.

[J7] A. Tewari*, J. Thies*, B. Mildenhall*, P. Srinivasan*, et al. Advances in Neural Rendering. *Computer Graphics Forum (EG STAR)*, 2022.

[J8] M. B R, A. Tewari, A. Dib, T. Weyrich, B. Bickel, H.-P. Seidel, H. Pfister, W. Matusik, L. Chevallier, M. Elgharib, and C. Theobalt. PhotoApp: Photorealistic Appearance Editing of Head Portraits. *ACM Transactions on Graphics (Proceedings SIGGRAPH)*, 2021.

[J9] A. Tewari, M. Elgharib, M. BR, F. Bernard, H.-P. Seidel, P. Pérez, M. Zöllhofer, and C. Theobalt. PIE: Portrait Image Embedding for Semantic Control. *ACM Transactions on Graphics (Proceedings SIGGRAPH Asia)*, 2020.

[J10] A. Tewari, M. Zollhöfer, F. Bernard, P. Garrido, H. Kim, P. Pérez, and C. Theobalt. High-Fidelity Monocular Face Reconstruction Based on an Unsupervised Model-Based Face Autoencoder. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2020.

[J11] A. Tewari*, O. Fried*, J. Thies*, et al. State of the Art on Neural Rendering. *Computer Graphics Forum (EG STAR 2020)*, 2020.

[J12] B. Egger, W. A. P. Smith, A. Tewari, et al. 3D Morphable Face Models - Past, Present and Future. *ACM Transactions on Graphics (Presented at SIGGRAPH)*, 2020.

[J13] M. Elgharib*, M. Mendiratta*, J. Thies, M. Nießner, H.-P. Seidel, A. Tewari, V. Golyanik, and C. Theobalt. Egocentric Videoconferencing. *ACM Transactions on Graphics (Proceedings SIGGRAPH Asia)*, 2020.

[J14] O. Fried, A. Tewari, M. Zollhöfer, A. Finkelstein, E. Shechtman, D. B. Goldman, K. Genova, Z. Jin, C. Theobalt, and M. Agrawala. Text-based Editing of Talking-head Video. *ACM Transactions on Graphics (Proceedings SIGGRAPH)*, 2019.

[J15] H. Kim, P. Garrido, A. Tewari, W. Xu, J. Thies, M. Nießner, P. Pérez, C. Richardt, M. Zollhöfer, and C. Theobalt. Deep Video Portraits. *ACM Transactions on Graphics (Proceedings SIGGRAPH)*, 2018.

Preprints and Manuscripts

* equal first authors † equal advising

[P1] J. Chen, J. Hu, J. Lasenby, and A. Tewari. Efficient Camera-Controlled Video Generation of Static Scenes via Sparse Diffusion and 3D Rendering, 2026. arXiv: 2601.09697 [cs.CV]. URL: https://arxiv.org/abs/2601.09697.

[P2] J. Hu, D. Olmeda Reino, and A. Tewari. S2PD: Serial-to-Parallel Diffusion for Physically and Logically Consistent Video Generation, 2026. arXiv: 2610.06847 [cs.CV]. URL: https://arxiv.org/abs/2610.06847.

[P3] Y. Yin, Z. Wen, S. Ye, J. Chen, Z. Zheng, N. Dai, H. Shi, A. Huang, Z. Zhang, A. Yuille, J. Xie, A. Tewari, and T. Shu. 3D-Belief: Embodied Belief Inference via Generative 3D World Modeling, 2026. arXiv: 2605.11367 [cs.CV]. URL: https://arxiv.org/abs/2605.11367.

Research Supervision

Current PhD students

2025–present Felix O'Mahony; co-supervised with Roberto Cipolla

2025–present Jieying Chen; co-supervised with Joan Lasenby

2025–present Jeffrey Hu

2026–present Ziyang Fu; co-supervised with Christian Theobalt at the MPI for Informatics

UROP and master's students

2026 Poulomi Dey, Ibrahim Suder, Bruno Hawkins, Haojun Yan, Daniel Korth

2025 Arsen Mikovic

Doctoral Thesis Examination

2026 Kirill Mazur, Imperial College London (external examiner).

2025 Fei Xu, University of Cambridge (internal examiner).

2025 Yassir Fathullah, University of Cambridge (internal examiner).

2025 Chenliang Zhou, University of Cambridge (internal examiner).

Teaching Experience

2025–2026 IIB project supervision, *University of Cambridge*.
Daniel Ciesla and Lauren Kwon

2026 GF4: Structure from Motion, *University of Cambridge*.
IIA project. 18 students

2026 Information Engineering (2P6), *St John's College, Cambridge*.
Small-group supervision

2026 Advanced Computer Vision (MLMI17), *University of Cambridge*.
Master's course

2025 World Models and Physical Intelligence, *Boston University*.
Guest lecture

2025 Probabilistic Machine Learning (4F13), *University of Cambridge*.
Fourth-year undergraduate course

2025 ELLIS Summer School on Probabilistic Machine Learning, *Cambridge*.
Lecture on generative models for 3D perception

2022–2023 Machine Learning for Inverse Graphics, *MIT, Cambridge, MA, USA*.
Guest lectures

2022 Neural Rendering, *Princeton University, virtual*.
Guest lecture

2021 Advances in Neural Rendering, *3DV and SIGGRAPH, virtual*.
Course tutor

2020 Neural Rendering, *CVPR and Eurographics, virtual*.
Tutorial tutor

2017–2021 Teaching Assistant, *Max Planck Institute for Informatics*, Saarbrücken, Germany.
Seminars:
- 3D Shape Analysis (2018)
- Computer Vision for Computer Graphics (2017, 2019, 2021)

2012–2013 Teaching Assistant, *International Institute of Information Technology*, Hyderabad, India.
Courses:
- Digital Signals Analysis and Applications (2013)
- Mathematics I (Discrete Mathematics) (2012, 2013)

Selected Invited Talks and Keynotes

2025–2026 Physical Inductive Biases for World Models.
Simon Fraser University; Beyond CAD Workshop, University of Edinburgh; University of Cambridge Wednesday Seminars; Waymo; SIGGRAPH 2026 Technical Committee Meeting.

2025 World Models and Physical Intelligence.
Eurographics Symposium on 3D Object Retrieval 2025 (keynote).

2025 Physical Inductive Biases.
Wayve, London; St John's College, Cambridge.

2024–2025 Learning to See the World in 3D.
Fourteen invited seminars, including Johns Hopkins, Cornell Tech, Michigan, UC San Diego, Stanford, UCLA and Caltech, TTIC, Northwestern, Columbia, Purdue, Maryland, Tübingen, Simon Fraser, and KAUST.

2023 3D Structured Generative Models.
Johns Hopkins; Carnegie Mellon; Meta Reality Labs Research; Princeton; MIT Graphics Seminar.

2023 Teaching AI to See the 3D World.
CSAIL and Imagination in Action: AI Frontiers & Implications.

2022 Finding 3D Structure in Unstructured 2D Data.
Rank Prize Symposium on Neural Rendering; Adobe Research; University of Oxford; Princeton University.

2022 Learning 3D Generative Models from 2D Data.
Dagstuhl Seminar on 3D Morphable Models and Beyond.

2021 Self-Supervised 3D Digitization of Faces.
MIT Vision and Graphics Seminar.

Academic Service

Editorial Associate Editor, *Computer Graphics Forum* (from 2025).

Leadership SIGGRAPH Technical Papers Committee (2026–2027); SIGGRAPH Asia Technical Papers Committee (2025–2026); ICLR Area Chair (2027); ICML Area Chair (2026); NeurIPS Area Chair (2025–2026); 3DV Programme Committee (2023, 2025).

Organisation SIGGRAPH Asia Workshop Committee Member (2026); SIGGRAPH Technical Papers Session Chair (2025–2026); SIGGRAPH Course on Advances in Neural Rendering (2021); 3DV Tutorial on Advances in Neural Rendering (2021); Eurographics Course on Neural Rendering (2020); CVPR Tutorial on Neural Rendering (2020).

Reviewing NSERC Discovery Grants; TPAMI; ACM TOG; IJCV; CVPR; ICCV; ECCV; NeurIPS; ICML; ICLR; SIGGRAPH and SIGGRAPH Asia; Eurographics; 3DV; BMVC.